

```

void sx126x_cal(void)
{
    check_busy();
    SPI_NSS_LOW();
    spi_rw(0x89);
    spi_rw(0x7f);
    SPI_NSS_HIGH();
}

power_temp = power_table[tx_power];

SetStandby(0); //0:STDBY_RC; 1:STDBY_XOSC
SetRegulatorMode();
SetPaConfig();

if(tcxo_flag)
{
    SetDIO3AsTCXOCtrl(DIO3_1_8V);
    sx126x_cal();
    ClearDeviceErrors();
}

SetDIO2AsRfSwitchCtrl();

SetPacketType(1); //0:GFSK; 1:LORA
SetRfFrequency(RF_freq); // RF_Freq = freq_reg*32M/(2^25)
SetTxParams(power_temp, SET_RAMP_10U); //set power and ramp_time
if(sensitivity_test)
{
    SetModulationParams(sf_temp, bw_temp, cr_temp, LDRO_OFF); //关低速率优化
}
else
{
    // ...
}

```

Adding this command after SetDIO3AsTCXOCtrl will enable normal startup

13.1.12 Calibrate Function

At power up the radio performs calibration of RC64k, RC13M, PLL and ADC. It is however possible to launch a calibration of one or several blocks at any time starting in STDBY_RC mode. The calibrate function starts the calibration of a block defined by *calibParam*.

Table 13-17: Calibrate SPI Transaction

Byte	0	1
Data from host	Opcode = 0x89	calibParam

The total calibration time if all blocks are calibrated is 3.5 ms. The calibration must be launched in STDBY_RC mode and the BUSY pins will be high during the calibration process. A falling edge of BUSY indicates the end of the procedure.

Table 13-18: Calibration Setting

CalibParam	Calibration Setting
Bit 0	0: RC64k calibration disabled 1: RC64k calibration enabled
Bit 1	0: RC13M calibration disabled 1: RC13M calibration enabled
Bit 2	0: PLL calibration disabled 1: PLL calibration enabled
Bit 3	0: ADC pulse calibration disabled 1: ADC pulse calibration enabled
Bit 4	0: ADC bulk N calibration disabled 1: ADC bulk N calibration enabled
Bit 5	0: ADC bulk P calibration disabled 1: ADC bulk P calibration enabled
Bit 6	0: Image calibration disabled 1: Image calibration enabled
Bit 7	0: RFU